

Computer Vision AI Architecture & Multi-Model Inference Pipeline

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1. Executive Overview & Problem Formulation

The Computer Vision perception core serves as the primary intelligence layer of the Mobile Urban Surveillance Platform. Operating directly on public transit buses, the subsystem ingests high-definition camera feeds and concurrently executes deep learning networks for traffic telemetry, road surface degradation assessment, infrastructure compliance auditing, and Automated Number Plate Recognition (ANPR).

Core Achievement: Zero-latency single-model cadenced execution delivering **28.4 FPS** on an NVIDIA GeForce RTX 3050 6GB Laptop GPU in FP16 Tensor Core precision, combining YOLO11s, ByteTrack, LiDAR shadow depth estimation, and EasyOCR consensus OCR without thread starvation.

2. Computer Vision Tech Stack & Framework Architecture

Framework / Engine	Version	Role in Platform	Architectural Rationale
PyTorch CUDA	2.2.0+cu121	Deep Learning Core	Native CUDA FP16 Tensor Core acceleration with minimal memory paging.
Ultralytics YOLO11s	8.1.0	Object Detection	C3k2 and SPPF architecture providing high recall for distant vehicles and pedestrians.
ByteTrack	Embedded	Trajectory Tracking	Kalman filter association maintaining persistent IDs through occlusions.
EasyOCR Ensemble	1.7.1	Plate Text Recognition	Multi-view consensus engine combining CLAHE, Otsu, and bilateral smoothing.
OpenCV (cv2)	4.8.1	Spatial Image Processing	Color-space segmentation (HSV), morphology kernels, Hough line detection.

3. Deep-Dive Model Components & Specifications

A. YOLO11s Traffic & ByteTrack Pedestrian Tracker

- **Decoupled Prediction Head:** Separates classification loss (BCE) from bounding box regression loss (CIoU + DFL).
- **Driver/Passenger Filtering:** Geometric centroid testing filters out passengers sitting inside vehicles, preventing false school zone pedestrian alarms.

B. Road Hazard Segmentation & LiDAR Depth Estimation (IRC:82-2015)

- **IRC Severity Tiers:** Evaluates potholes as P1 Critical ($>0.08\text{m}^2$ / 24h SLA), P2 High ($0.03\text{-}0.08\text{m}^2$ / 48h SLA), or P3 Medium ($<0.03\text{m}^2$ / 7d SLA).
- **Photometric Shadow Gradient Depth:** Models depression depth via perspective camera geometry and cavity shadow luminance attenuation.

C. Road Infrastructure Compliance Engine (IRC:119-2015 & IRC:35-2015)

- **Unpaved Dirt Road False-Positive Suppression:** Evaluates rural soil chromatic content in HSV space (H: 14..36). If sandy soil > 40%, median divider detection is fully suppressed.
- **Thermoplastic Crosswalk Wear:** Directional Sobel edge analysis detects degraded or missing pedestrian zebra crossings.
- **Regulatory Signboard Deflection:** Polygonal contour analysis detects damaged or tilted traffic signboards ($|\theta| > 12^\circ$).

D. MoRTH Indian Registration Plate ANPR Consensus Engine

- **Multi-View Ensemble:** Processes plates across CLAHE, Otsu binarization, and bilateral smoothing to ensure readability under motion blur and rain glare.
- **MoRTH Synthesizer:** Reconstructs degraded OCR strings into valid Indian vehicle registration format: SS-DD-LL-NNNN (e.g. TN-09-AB-1234).